AAE 41200
Introduction to Computational Fluid Dynamics (CFD)
Fall 2025

(Face-to-Face, CRN 10053, 3 cr hr)

Prerequisite: AAE 334 (Aerodynamics); or permission of instructor

Instructor

Prof. Gregory A. Blaisdell

Teaching Assistant

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Course Description

From the Purdue Course Catalog: "Introductory course in the formulation and application of finite difference methods for solving fluid flow problems. Classification of partial differential equations and formulation of well-posed problems. Discrete approximation of partial differential equations: stability, consistency, and convergence. Finite-volume formulations. Survey of methods for solving hyperbolic, elliptic, and parabolic problems. Formulation of discrete boundary conditions. Application of methods to one- and two-dimensional flow problems."

Course Topics

- Interpolation, numerical integration, Richardson extrapolation, finite difference approximations of derivatives, order of accuracy, evaluating numerical solutions by comparing to known solutions and grid resolution studies
- Application of the finite volume method to 1-D steady diffusion, Thomas algorithm for tridiagonal linear systems
- Application of the finite volume method to 1-D steady convection-diffusion, non-uniform grids, unsteady convection and diffusion, issues of stability
- Application of the finite volume method to 2-D problems, iterative solution of linear systems, grid generation, generalized coordinates, visualization of 2-D fields, application to 2-D potential flow over an airfoil
- Formulation of pressure-based solvers for the Navier-Stokes equations
- Formulation of density-based solvers for the Navier-Stokes equations
- Use of ANSYS Fluent for solution of steady and unsteady, 2-D and 3-D problems
- Formulation and limitations of turbulence models

Course Learning Outcomes

During this semester students are expected to learn to following:

- Learn the basics of finite volume approximations of Partial Differential Equations, including error estimation, on logically rectangular grids in one and two dimensions
- Learn how to derive and use interpolation and finite difference formulas
- Assess the accuracy of a numerical solutions by comparison to known solutions of simple test problems and by mesh refinement studies
- Learn how CFD is used to predict quantities of engineering interest, such as forces on airfoils or thrust from nozzles
- Learn how to run the ANSYS/Fluent commercial CFD code for relatively simple problems
- Learn how to present computational results in a useful fashion.

Textbook (recommended)

An Introduction to Computational Fluid Dynamics – The Finite Volume Method (2nd ed.)
H. K. Versteeg and W. Malalasekera
Pearson Education 2007, ISBN 9780131274983
(Available at https://www.amazon.com/Introduction-Computational-Fluid-Dynamics-Finite/dp/0131274988)

References

Fundamentals of Engineering Numerical Analysis (Second edition)
Parviz Moin
Cambridge University Press 2010

Computational Fluid Dynamics: The Basics with Applications
John D. Anderson, Jr.
McGraw-Hill 1995

Computational Fluid Mechanics and Heat Transfer (Fourth edition)
Dale A. Anderson, John C. Tannehill, Richard H. Pletcher and Ramakanth Munipalli
CRC Press 2020

Course Material

Material for this course will be posted on Brightspace (login required):
https://purdue.brightspace.com/d2l/home/1353484

Course Notes

The lecture notes will be made available on Brightspace. Except for Lecture Notes 1, most of the lecture notes have blank spaces where information will be added during class. Students are responsible for

printing or downloading the course notes before lecture so they can add material to the printed notes during class. If you miss class, please get the missing material for the notes from another student.

The course notes contain copyrighted material. Therefore, do not post the files on the Internet or share them with anyone not enrolled in this course. This includes a prohibition against uploading course material to AI systems.

Grading

The basis for the grade for this course will be scores on the homework and project. There will not be any exams. Because there will be a small number of homework assignments (7), each one will have a significant impact on your course grade (10%). Therefore, it is very important that all the homework assignments be turned in. Even if they are incomplete, it is better to get partial credit than to earn a zero.

The project will be worth the equivalent of 30% of the course grade. If your homework scores are too low (typically below 60%, but that is a decision I will make later), you will not be assigned to work with a team for the project. If you are not assigned to a team, you will have the option of doing the project on your own with a reduced scope of work. The purpose of this policy is to ensure all students on a team are prepared to contribute effectively.

I expect to use a grading scheme of A = 90-100%, B = 80-90%, C = 70-80%, D = 60-70%, F < 60%. I will use a +/- grading scheme with the lower portion (to be determined by me at the end of the semester) of each range earning a "-" grade and the top portion earning a "+" grade. I may choose to make the grading scheme more lenient than this, but it will not be stricter than this.

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course Brightspace webpage or can be obtained by contacting the instructors or TA via email. You are expected to read your @purdue.edu email on a frequent basis. Here are ways to get information about changes in this course: Brightspace and my email address ([email address]). (Piazza.com is a secondary source of information.)

Additional Links

- Student edition of ANSYS Workbench (free): https://www.ansys.com/academic/students/ansys-student. (*Note that the current student edition is version 2025 R2, while the version available in the AAE computer lab is 2025 R1. Workbench projects saved from a newer version cannot be opened with earlier versions of the software. Therefore, if you install ANSYS on your own computer, you should download and install version 2025 R1 to avoid compatibility issues when opening projects on the AAE computers or sharing your project with students on your team. Earlier versions of ANSYS are available by expanding the Prior Releases (+) tab*.)
- The ANSYS-Fluent tutorials:
  https://confluence.cornell.edu/display/SIMULATION/FLUENT+Learning+Modules
  and
  https://courses.ansys.com/index.php/courses/simcafe-fluids/
  (plus resources on YouTube.com)

Previous students suggested these resources:

- Basic heat transfer concepts: https://web2.clarkson.edu/projects/subramanian/ch330/notes/Introduction%20to%20Heat%20Transfer.pdf
- Control volumes and conservation laws: https://www.stat.uchicago.edu/~guillaumebal/COURSES/E3102/MSpieg-Conserveq.pdf

Homework Assignments and Project

Homework assignments will be given out at various times (see course schedule). Generally, the homework assignments will consist of two parts. Some questions will be related to the theory presented in class or the textbook and may involve modifying and running small Fortran or Matlab programs. The other part of the homework will require you to work through tutorials to learn to use ANSYS Workbench and to use the software to go beyond what is shown in the tutorials.

The project will be of longer duration and more involved than the homework assignments. The project will involve running the ANSYS Fluent commercial CFD software package. This will be a team project. I will assign the members of each team. You will be required to write a project report, and your team will make a video with an oral presentation on your project. You will also have to review two presentations from other teams. More information on the project and the requirements for the report, presentation and review will be given out later.

Uploading Homework

You must upload homework to the class account on Gradescope.com. Upload your homework as a single PDF file. Examine your PDF file before submitting it to ensure that it is legible. If the TA or grader cannot read what you have written, then it will be considered incorrect. Also, you are responsible for correctly assigning the various problems or parts of problems to the pages of your solution. Failure to correctly assign pages of your solution to the corresponding problem parts will result in no credit for that part of the homework. (The Adobe Scan app works very well for scanning homework and is free.)

Late Homework

The nature of computer programming is such that getting programs to work correctly usually takes more time than you initially anticipate. A program may be 99.9% correct and still not produce useful results. Therefore, it is important that you start homework early. However, since there may be times that you cannot debug your program in time to get results and finish the write-up before the due date, I will make some allowances for late homework. You may turn in homework assignments by the late due date for reduced credit. Typically, the late due date will be the next day on which the course meets after the original due date, e.g., Monday after a Friday due date, Wednesday after a Monday due date, or Friday after a Wednesday due date. Homework will be due at 11:59 pm on the day it is due. The late penalty will be 10% of the total value of the assignment (e.g., -2 pts out of 20). Homework may not be turned in after that time (other than by special arrangement for extraordinary circumstances). Keep in mind that all homework problems must be turned in at the same time – they are all either on time or late. The late

homework policy does not extend to the project. The project may not be turned in late (except by special arrangement for extraordinary circumstances).

Collaboration

With the exception of the group project, the work done for this class is to be done individually. You are encouraged to get help from other students, if needed. However, the work you turn in must be your own. The computer programs you turn in must represent your own work, the results you show must come from your own program (unless a program is provided to the class for the assignment), and the write-up or report must be your own work. If you have difficulty, please seek help — from me, the TA, or other students — but do not just copy from others. Changing variable names and rewording comments taken from another person's code or write-up is still copying. Copying computer code, results or portions of a write-up from others constitutes cheating, and it will be dealt with as such. This includes copying from solutions of the same or similar problems from previous years.

In addition, students are reminded that the homework assignments and solutions are copyrighted material. They may not be posted on the Internet or given to others. Specifically, they may not be bartered or uploaded to Course Hero or similar websites. Also, they may not be entered into AI systems.

Use of Artificial Intelligence (AI)

Use of Artificial Intelligence or Large Language Models (AI/LLM) to generate parts of a homework solution or parts of the team project report or video presentation will be considered a blatant attempt at cheating and will result in a failing grade in the course. This prohibition does not extend to the use of software systems to correct your spelling or grammar.